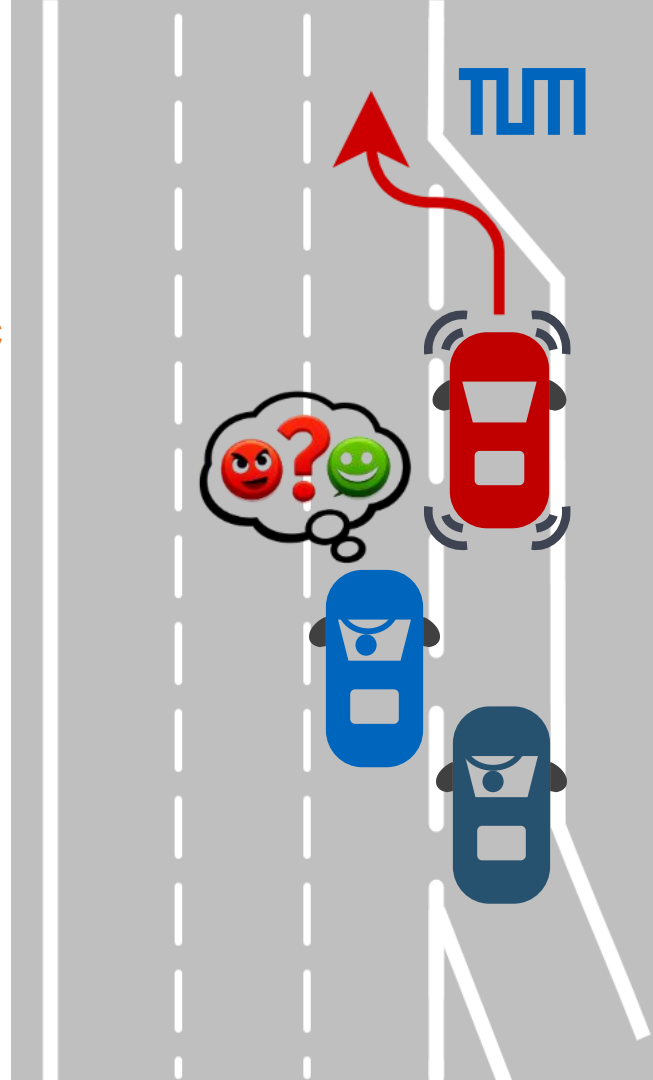


Modeling Driver Interaction Behavior in Mixed Traffic based on Script and Game Theory

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Theoretical background

(Tomkins, 1978) (Schank and Abelson, 1977)

Script Theory

“

Script theory is a cognitive model that explains how people organize knowledge and memory through structured sequences of actions called "**scripts**".

”

- ▶ Learn scripts through repetitive social interaction
- ▶ Predict, interpret, and understand new social experiences throughout life
- ▶ Help us act efficiently in familiar contexts
- ▶ e.g., a script for going out to eat (enter → order → eat → leave)

Game Theory

(Nash, 1950) (von Stackelberg, 1977)



Rock, Scissors, Paper



Auction



Chess

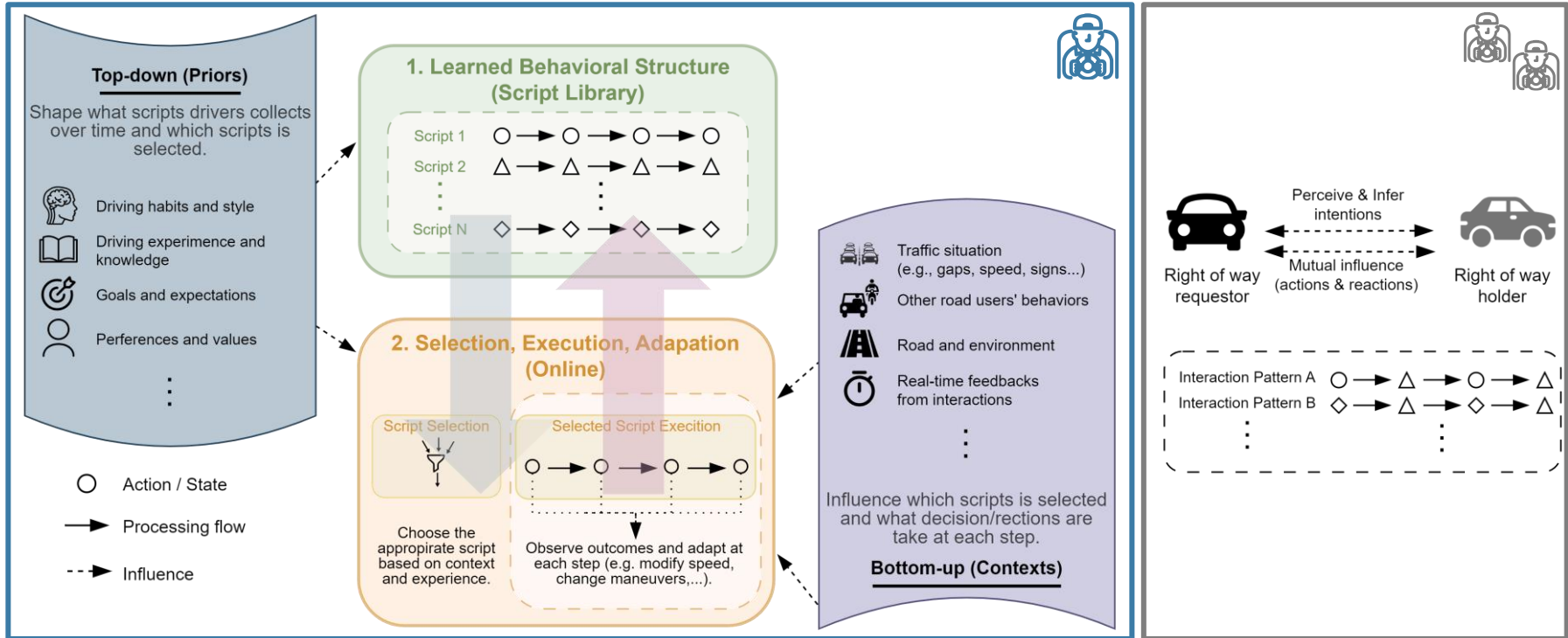


Poker

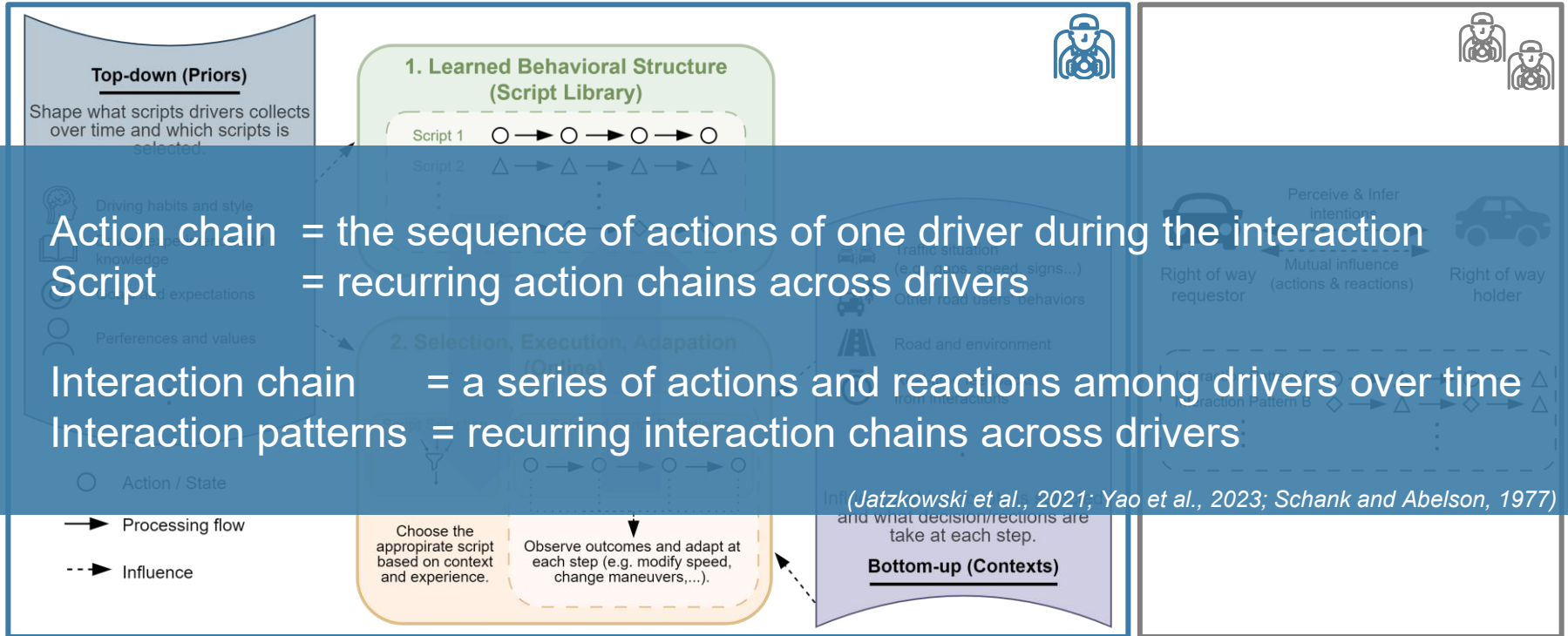
Core Idea

- ▶ A game exists when one person's outcome depends on both their own choices and others' choices.
- ▶ Three basic elements: Players, Strategies, Payoff.
- ▶ Players aim to maximize their payoffs and are self-interested.
- ▶ Players consider the responses/reactions of other players.

Driver Interaction Behavior



Driver Interaction Behavior



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